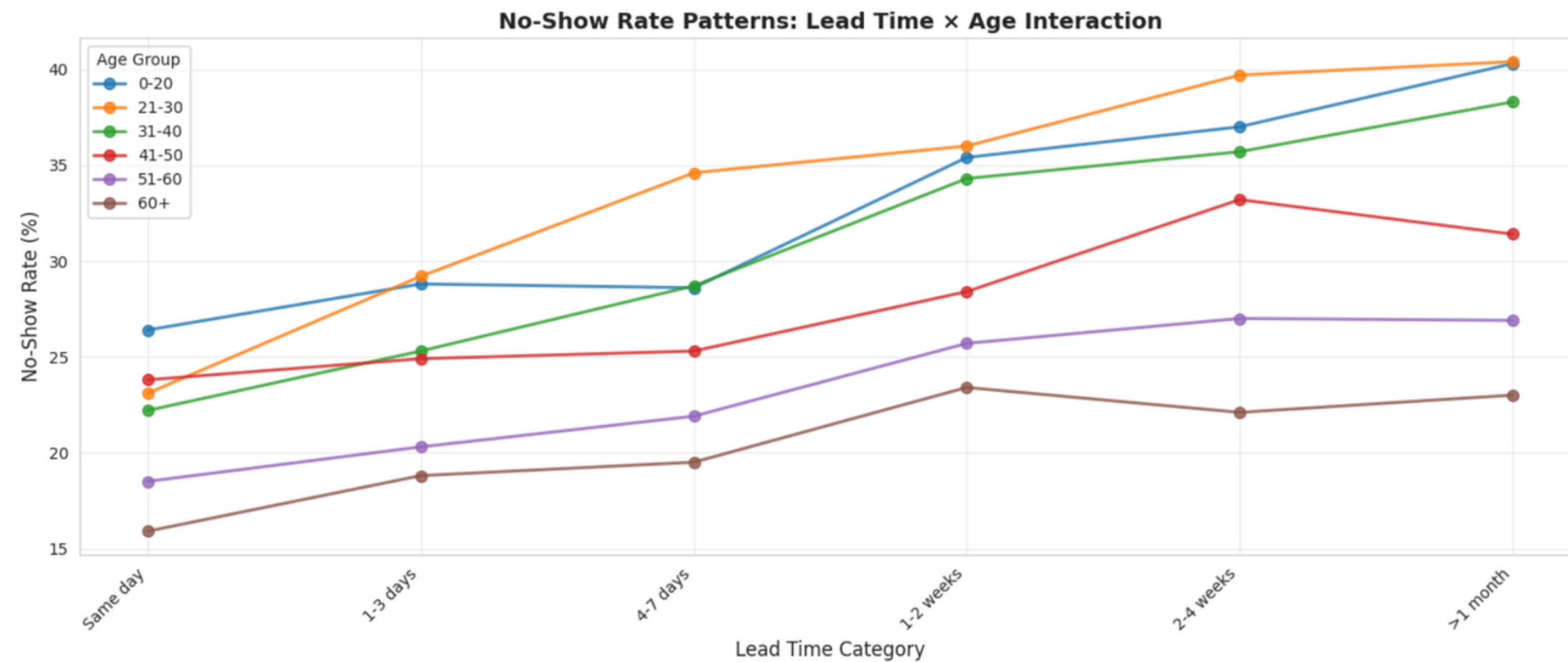




# Predictive Overbooking Optimization for Reducing Clinic No-Shows

Fall 2025

CPE 608: Applied Modelling and Optimization



# Problem Description

## The Challenge: Patient No-Shows in Healthcare, a \$150 Billion Problem<sup>1</sup>

~20-30%<sup>2</sup> of scheduled medical appointments result in no-shows, causing:

### Financial Impact

- Wasted physician time and clinic resources
- Lost revenue (~\$150+ per unused slot)
- Increased operating costs

### Access to Care

- Blocked appointment slots that other patients needed
- Longer wait times for all patients
- Reduced overall healthcare capacity

### Clinical Outcomes

- Delayed treatment for patients who miss appointments
- Potential worsening of health conditions
- Disrupted continuity of care

#### The Dilemma:

Book exactly at capacity → Waste slots when patients don't show  
Overbook aggressively → Overflow causes long waits, patient dissatisfaction

 **Our Objective:** Find the optimal overbooking policy that balances clinic efficiency against patient experience using predictive modeling and optimization

# Optimization Model - Formulation

## Decision Variable:

$k$  = number of extra patients to book beyond capacity  $C$

## Objective Function:

$$\min_k E[\text{Total Cost}(k)] = E[\text{Idle Cost}] + E[\text{Overflow Cost}]$$

Where:

Idle Cost =  $c_{\text{idle}} \times \max(0, C - S)$  [unused slots]

Overflow Cost =  $c_{\text{overflow}} \times \max(0, S - C)$  [excess patients]

$S$  = number of shows  $\sim$  Poisson-Binomial( $1-p_1, 1-p_2, \dots, 1-p_n$ )

$p_i$  = predicted no-show probability from ML model

## Cost Parameters:

$c_{\text{idle}}$  = \$150/slot (physician idle time)

$c_{\text{overflow}}$  = \$75/patient (patient waiting/rescheduling)

$C$  = 20 patients/session

# Optimization Model - Constraints and Rationale

## Constraints:

1.  $k \geq 0$  (non-negativity)
2.  $k \leq C$  (cannot book more than 2× capacity)
3.  $P(\text{overflow}) \leq \alpha$  (optional service level constraint)

## Why This Formulation?

Captures the core tradeoff:

- Under-booking → wasted physician time (\$150/slot)
- Over-booking → patient overflow (\$75/patient)

Uses patient-specific predictions:

- Each patient has individual  $p_i$  from ML model
- Enables slot-specific overbooking (adaptive to risk composition)

Asymmetric costs reflect reality:

- Idle time is more expensive than overflow
- Optimal policy should lean toward overbooking

# Proposed Solution - Convexity

The expected cost function  $E[\text{Cost}(k)]$  is convex in  $k$ .

## Proof:

- Idle cost:  $E[\max(0, C-S)]$  is convex and decreasing in  $k$
- Overflow cost:  $E[\max(0, S-C)]$  is convex and increasing in  $k$
- Sum of convex functions is convex

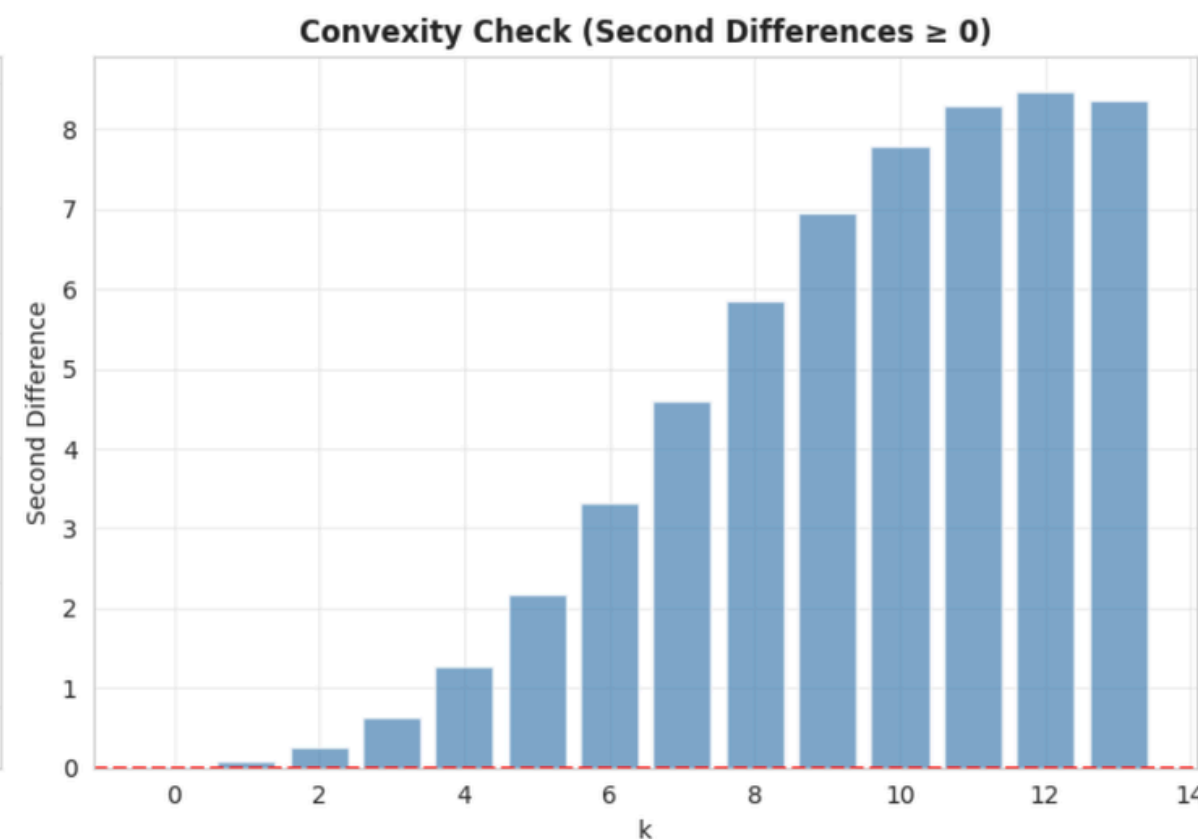
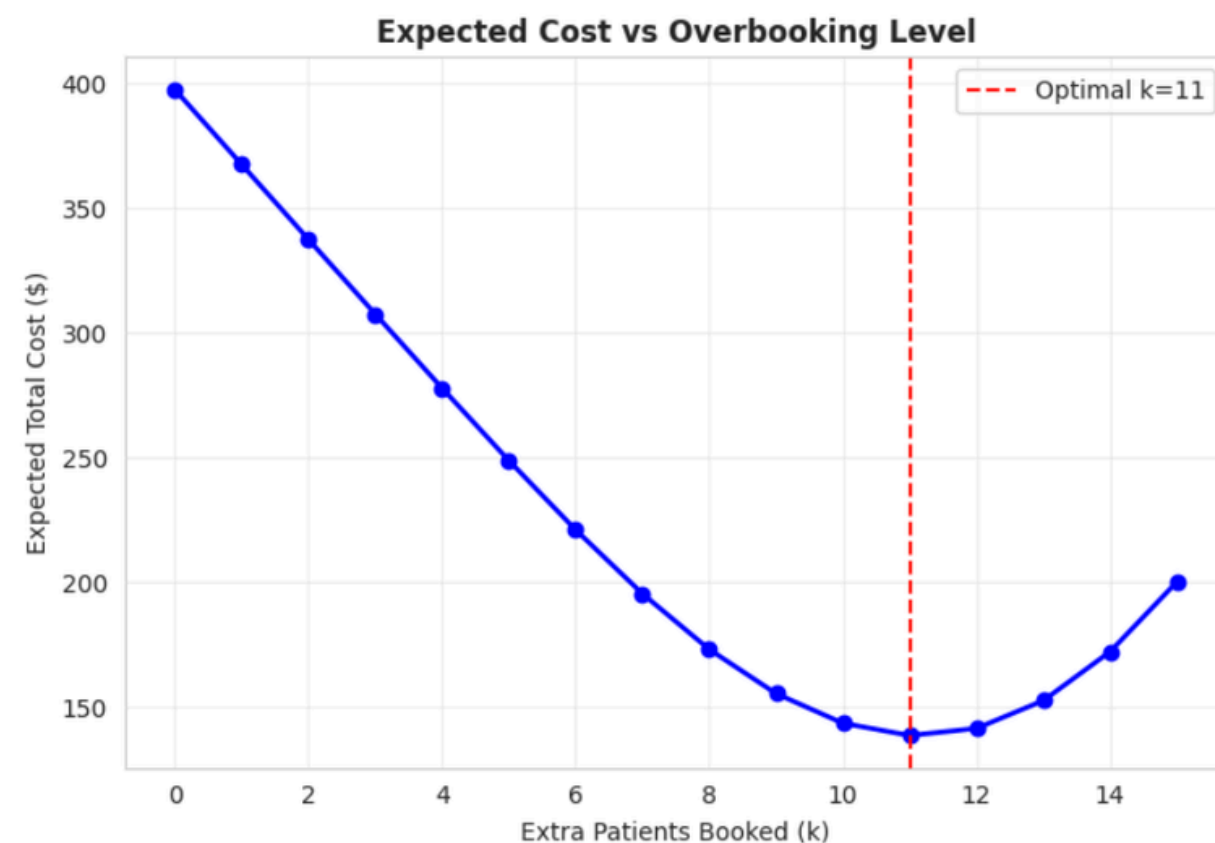
## Verified Numerically:

- Second differences  $\geq 0$  for all  $k \in \{0, 1, \dots, 15\}$

## Implication:

- A unique global minimum  $k^*$  exists
- No local minima traps
- Simple enumeration finds optimal solution

Cost function is convex: True  
Optimal  $k$  for sample slot: 11





# Proposed Solution - Feasible Set and Solution Method

## Feasible Set:

$k \in \{0, 1, 2, \dots, 20\}$  (discrete, finite)

This small search space enables:

- Complete enumeration of all possible  $k$  values
- Monte Carlo estimation of  $E[\text{Cost}(k)]$  for each  $k$
- Guaranteed finding of global optimum

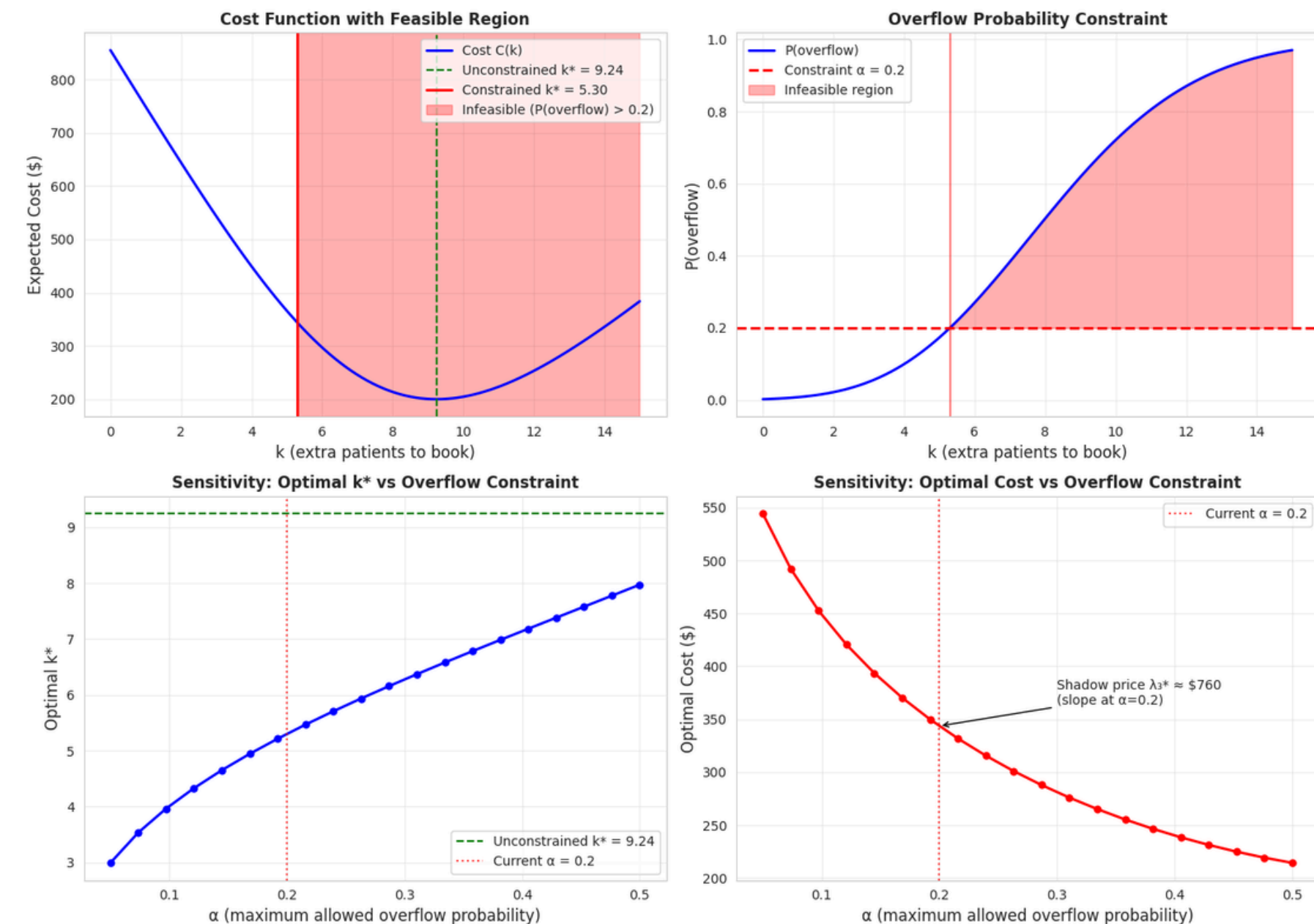
## Solution Approach:

1. Use XGBoost to predict  $p_i$  for each patient in slot
2. For each candidate  $k \in \{0, 1, \dots, 20\}$ :
  - Run  $N=1000$  Monte Carlo simulations
  - Estimate  $E[\text{Cost}(k)]$  with 95% confidence intervals
3. Select  $k^* = \text{argmin } E[\text{Cost}(k)]$

## Closed-Form Approximation (Newsvendor):

$$k^* \approx F^{-1}(c_{\text{idle}} / (c_{\text{idle}} + c_{\text{overflow}}))$$

where  $F$  is the CDF of the Poisson-Binomial distribution





# Proposed Solution - Results

**Optimal Policy:** cost\_optimal achieves  $k^* \approx 10$  extra patients

## Key Results:

- Mean cost: \$457/slot [95% CI: \$451, \$463]
- Savings vs no overbooking: 64.5% (\$832/slot)
- Utilization: 85.3% (vs 57% without overbooking)
- Overflow probability: 11.5%
- Projected annual savings: ~\$598,000 (across 719 slots)

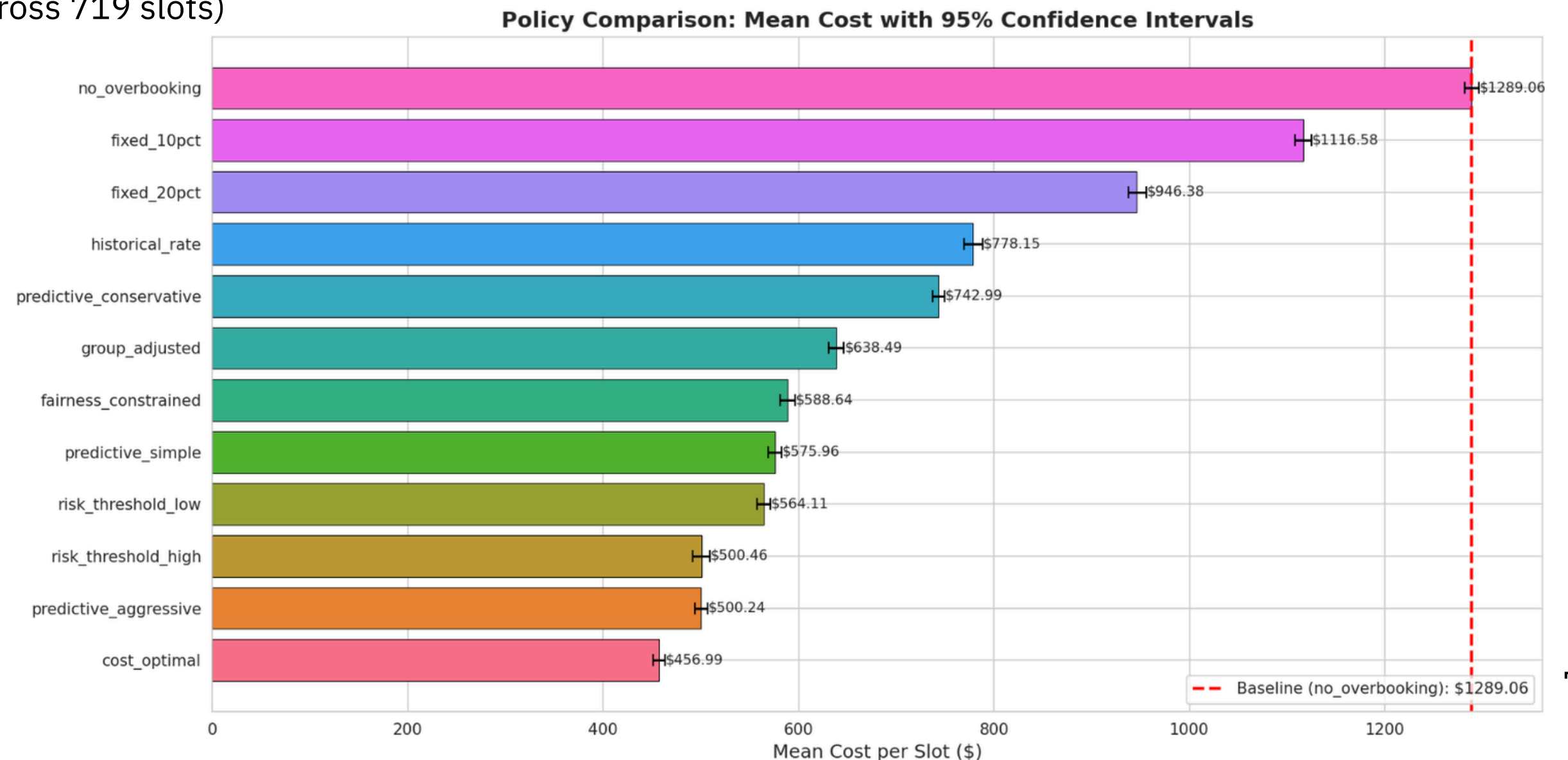
## Statistical Significance:

- $p < 0.0001$  vs baseline
- Robust across 8 stress scenarios

**Gap:** cost\_optimal saves \$832/slot vs no\_overbooking That's a 2.8x cost reduction for the same clinic capacity.

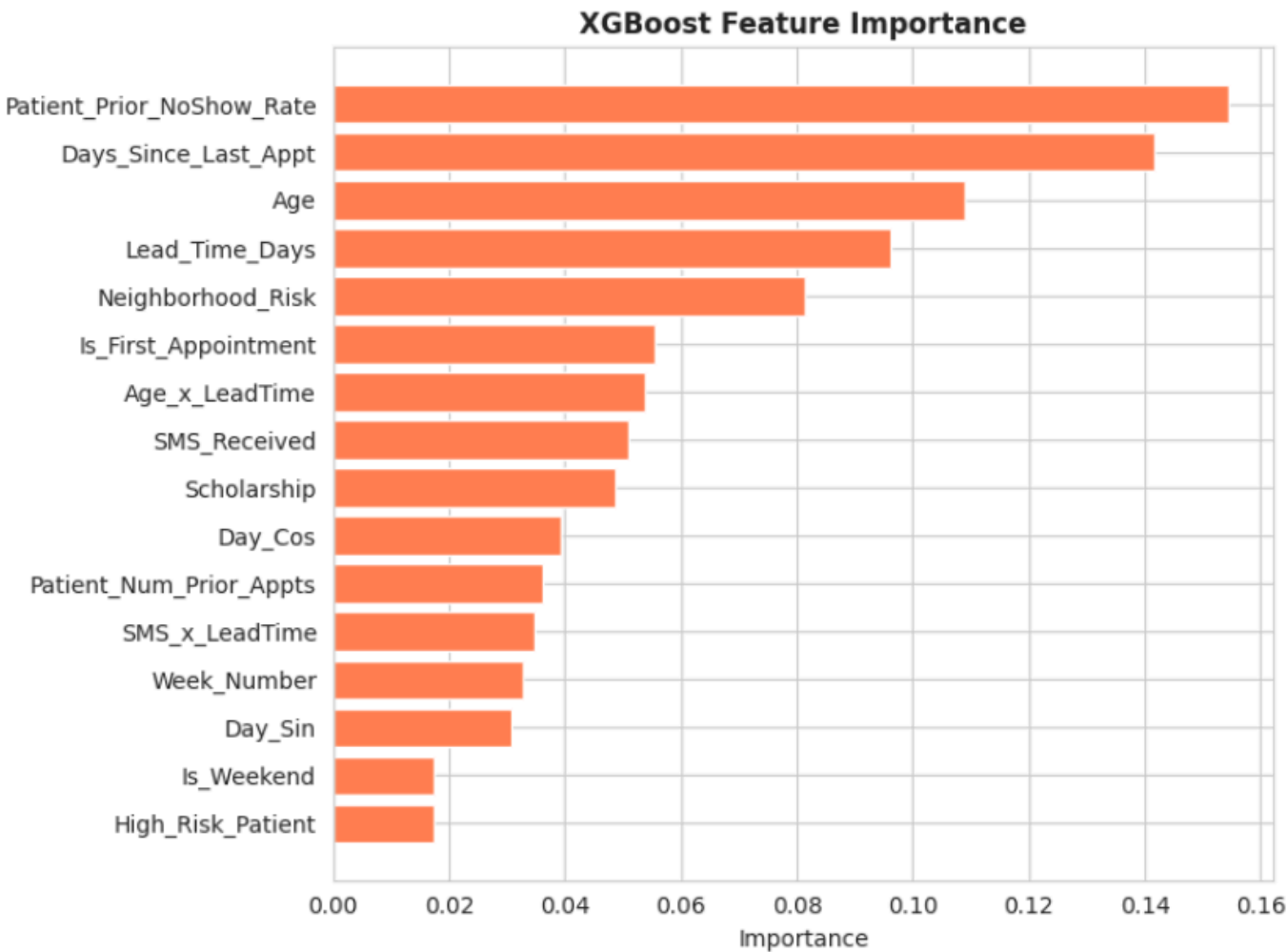
**Note:** cost\_optimal achieves lower cost but has 11.5% overflow probability. For patient centric clinics, predictive\_simple (\$576, 5.3% overflow) may be preferred.

**Insight:** ML-based policies (green/orange bars) consistently outperform fixed-rate policies (pink/purple bars) — demonstrating the value of patient-specific predictions over one-size-fits-all rules.



# Validating Solution - Dataset and Experimental Setup

| Component           | Details                                                       |
|---------------------|---------------------------------------------------------------|
| Dataset             | Brazilian Healthcare No-Show Appointments (Kaggle)            |
| Size                | 14,392 appointments across 719 clinic slots                   |
| Slot Size           | 20 patients per session                                       |
| Target Variable     | No-show (binary: 0/1)                                         |
| Actual No-Show Rate | 26.10%                                                        |
| Key Features        | Age, Lead Time, Prior No-Show Rate, Scholarship, SMS Reminder |



The top 3 predictors (Prior No-Show Rate, Days Since Last Appt, Age) are behavioral/temporal; suggesting no-shows are habitual, not random.



# Validating Solution - Theoretical Validation (Optimization Framework)

At  $k^* = 5.2994$ :

## 1. PRIMAL FEASIBILITY:

$$\begin{aligned} g_1(k^*) &= -k^* = -5.2994 \leq 0 \quad \checkmark \\ g_2(k^*) &= k^* - k_{\max} = -9.7006 \leq 0 \quad \checkmark \\ g_3(k^*) &= P(\text{overflow}) - \alpha = 0.0000 \leq 0 \quad \checkmark \end{aligned}$$

## 2. ACTIVE CONSTRAINTS:

$$\begin{aligned} g_1 (k \geq 0) &\text{ active: False } (k^* = 5.2994) \\ g_2 (k \leq k_{\max}) &\text{ active: False } (k^* - k_{\max} = -9.7006) \\ g_3 (P(\text{overflow}) \leq \alpha) &\text{ active: True } (P = 0.2000, \alpha = 0.2) \end{aligned}$$

## 3. SHADOW PRICES (Dual Variables):

$$\begin{aligned} \lambda_1^* \text{ (for } k \geq 0\text{)} &: 0 \text{ (inactive)} \\ \lambda_2^* \text{ (for } k \leq k_{\max}\text{)} &: 0 \text{ (inactive)} \\ \lambda_3^* \text{ (for } P(\text{overflow}) \leq \alpha\text{)} &: \approx \$759.80 \text{ per 1\% overflow allowance} \end{aligned}$$

## 4. STATIONARITY:

$$\begin{aligned} \partial C / \partial k &= -72.2484 \\ \partial P(\text{overflow}) / \partial k &= 0.092817 \\ \text{From stationarity: } \lambda_3^* &= -\partial C / \partial k / \partial P(\text{overflow}) / \partial k = 778.39 \\ \text{Verification: } \partial C / \partial k + \lambda_3^* \cdot \partial P / \partial k &= 0.000000 \approx 0 \quad \checkmark \end{aligned}$$

## 5. COMPLEMENTARY SLACKNESS:

$$\begin{aligned} \lambda_1^* \cdot g_1(k^*) &= 0 \cdot -5.2994 = 0 \quad \checkmark \\ \lambda_2^* \cdot g_2(k^*) &= 0 \cdot -9.7006 = 0 \quad \checkmark \\ \lambda_3^* \cdot g_3(k^*) &= 759.80 \cdot 0.000000 \approx 0 \quad \checkmark \text{ (constraint binding)} \end{aligned}$$

**Convexity Proof:** Cost function  $C(k)$  is convex in overbooking level  $k$

- Second differences  $\geq 0$  across all  $k$  values
- Guarantees unique global optimum

**KKT Conditions Verified:**

- Stationarity:  $\partial C / \partial k + \lambda_3 \cdot \partial P(\text{overflow}) / \partial k = 0 \quad \checkmark$
- Primal feasibility: All constraints satisfied  $\checkmark$
- Complementary slackness:  $\lambda \cdot g(k^*) = 0 \quad \checkmark$

**Shadow Price:**  $\lambda_3^* \approx \$760$  per 1% change in overflow constraint

- Quantifies cost-service level tradeoff

# Validating Solution - Simulation Engine Validation

## Deterministic Tests (All Passed):

| Test   | Validation                             |
|--------|----------------------------------------|
| Test 1 | no_overbooking correctly returns k=0 ✓ |
| Test 2 | fixed_10pct correctly returns k=2 ✓    |
| Test 3 | Cost = \$0 when shows = capacity ✓     |
| Test 4 | Overflow cost calculation correct ✓    |
| Test 5 | Idle cost calculation correct ✓        |

## Statistical Validation (Law of Large Numbers):

- Theoretical no-show rate: 0.4295
- Simulated no-show rate (N=1000): 0.4296
- Difference: < 0.001 ✓
- Confirms simulation correctly samples from predicted probabilities

running deterministic validation tests

test 1 passed – no\_overbooking correctly returns k=0

test 2 passed – no\_overbooking correctly returns k=0

test 3 passed – cost is \$0 when simulated shows exactly match capacity

test 4 passed – cost function overflow calculation is correct

test 5 passed – cost function idle calculation is correct

running statistical validation – law of large numbers test

Running simulation for policy predictive\_simple with N=1000 iterations

Running simulation for policy no\_overbooking with N=1000 iterations

theoretical no show rate (avg p(no-show) for all 14832 patients): 0.429460064795048

simulated no-show rate (no\_overbooking policy, N=1000): 0.4295920723226704

statistical test passed and simulated no show rate converges to the expected rate (tolerance: 0.005)

# Validating Solution - Monte Carlo Simulation

| Parameter                 | Value            |
|---------------------------|------------------|
| Policies tested           | 12               |
| Iterations per policy     | 1,000            |
| Slots per iteration       | 719              |
| <i>Total observations</i> | <i>8,628,000</i> |

### Stress Testing:

- 8 scenarios × 5 key policies × 500 iterations
- Scenarios: baseline, ±30% no-shows, ±50% no-shows, 10-20% model noise

### Cost Sensitivity:

- 5 cost configurations tested (ratio c\_idle/c\_overflow from 0.25 to 4.0)

**Why 8.6M observations?** Ensures tight confidence intervals (±\$6 per slot) and statistically significant comparisons between policies.

**Purpose:** Simulate real-world uncertainty; patient no-shows are stochastic, so we test each policy across thousands of possible outcomes.

**Validation:** Simulated no-show rate (42.96%) matched theoretical prediction (42.95%), confirming simulation fidelity.

```

RUNNING MONTE CARLO SIMULATIONS (N=1000)
=====

Running no_overbooking with N=1000 iterations...
  ✓ no_overbooking: 719000 observations
Running fixed_10pct with N=1000 iterations...
  ✓ fixed_10pct: 719000 observations
Running fixed_20pct with N=1000 iterations...
  ✓ fixed_20pct: 719000 observations
Running historical_rate with N=1000 iterations...
  ✓ historical_rate: 719000 observations
Running predictive_simple with N=1000 iterations...
  ✓ predictive_simple: 719000 observations
Running predictive_conservative with N=1000 iterations...
  ✓ predictive_conservative: 719000 observations
Running predictive_aggressive with N=1000 iterations...
  ✓ predictive_aggressive: 719000 observations
Running cost_optimal with N=1000 iterations...
  ✓ cost_optimal: 719000 observations
Running risk_threshold_low with N=1000 iterations...
  ✓ risk_threshold_low: 719000 observations
Running risk_threshold_high with N=1000 iterations...
  ✓ risk_threshold_high: 719000 observations
Running fairness_constrained with N=1000 iterations...
  ✓ fairness_constrained: 719000 observations
Running group_adjusted with N=1000 iterations...
  ✓ group_adjusted: 719000 observations

✓ All simulations complete!
Total observations: 8,628,000

```



# Validating Solution - Statistical Rigor

### Bootstrap Confidence Intervals:

- 95% CI computed via 1,000 bootstrap resamples
- Example: cost\_optimal mean = \$456.99 [451.39, 462.96]

### Significance Testing:

- Paired bootstrap tests: All policies vs baseline
- All 11 policies significantly better than no\_overbooking ( $p < 0.0001$ )

| SIGNIFICANCE TESTS VS BASELINE (no_overbooking) |                                     |              |
|-------------------------------------------------|-------------------------------------|--------------|
| fixed_10pct                                     | : diff=\$-172.48 [-174.40, -170.52] | p=0.0000 *** |
| fixed_20pct                                     | : diff=\$-342.69 [-345.54, -339.87] | p=0.0000 *** |
| historical_rate                                 | : diff=\$-510.91 [-514.26, -507.53] | p=0.0000 *** |
| predictive_simple                               | : diff=\$-713.11 [-716.68, -709.50] | p=0.0000 *** |
| predictive_conservative                         | : diff=\$-546.07 [-549.91, -542.21] | p=0.0000 *** |
| predictive_aggressive                           | : diff=\$-788.82 [-792.18, -785.41] | p=0.0000 *** |
| cost_optimal                                    | : diff=\$-832.07 [-835.67, -828.52] | p=0.0000 *** |
| risk_threshold_low                              | : diff=\$-724.95 [-734.17, -715.65] | p=0.0000 *** |
| risk_threshold_high                             | : diff=\$-788.60 [-792.50, -784.53] | p=0.0000 *** |
| fairness_constrained                            | : diff=\$-700.42 [-704.94, -695.84] | p=0.0000 *** |
| group_adjusted                                  | : diff=\$-650.57 [-654.42, -646.57] | p=0.0000 *** |

### Pairwise Comparisons (Top 5):

| Comparison                             | Difference | p-value     |
|----------------------------------------|------------|-------------|
| cost_optimal vs predictive_aggressive  | -\$43.25   | <0.0001***  |
| cost_optimal vs predictive_simple      | -\$118.97  | <0.0001***  |
| pred_aggressive vs risk_threshold_high | -\$0.22    | 0.9038 (ns) |

| PAIRWISE TESTS: TOP 5 POLICIES                                                                                              |                          |                                  |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------|
| Top 5 policies: ['cost_optimal', 'predictive_aggressive', 'risk_threshold_high', 'risk_threshold_low', 'predictive_simple'] |                          |                                  |
| cost_optimal                                                                                                                | vs predictive_aggressive | : diff=\$-43.254 p=0.0000 [***]  |
| cost_optimal                                                                                                                | vs risk_threshold_high   | : diff=\$-43.474 p=0.0000 [***]  |
| cost_optimal                                                                                                                | vs risk_threshold_low    | : diff=\$-107.122 p=0.0000 [***] |
| cost_optimal                                                                                                                | vs predictive_simple     | : diff=\$-118.968 p=0.0000 [***] |
| predictive_aggressive                                                                                                       | vs risk_threshold_high   | : diff=\$-0.220 p=0.9038 [ns]    |
| predictive_aggressive                                                                                                       | vs risk_threshold_low    | : diff=\$-63.868 p=0.0000 [***]  |
| predictive_aggressive                                                                                                       | vs predictive_simple     | : diff=\$-75.714 p=0.0000 [***]  |
| risk_threshold_high                                                                                                         | vs risk_threshold_low    | : diff=\$-63.648 p=0.0000 [***]  |
| risk_threshold_high                                                                                                         | vs predictive_simple     | : diff=\$-75.494 p=0.0000 [***]  |
| risk_threshold_low                                                                                                          | vs predictive_simple     | : diff=\$-11.846 p=0.0036 [***]  |

# Key Insights - Policy Performance & Cost Sensitivity

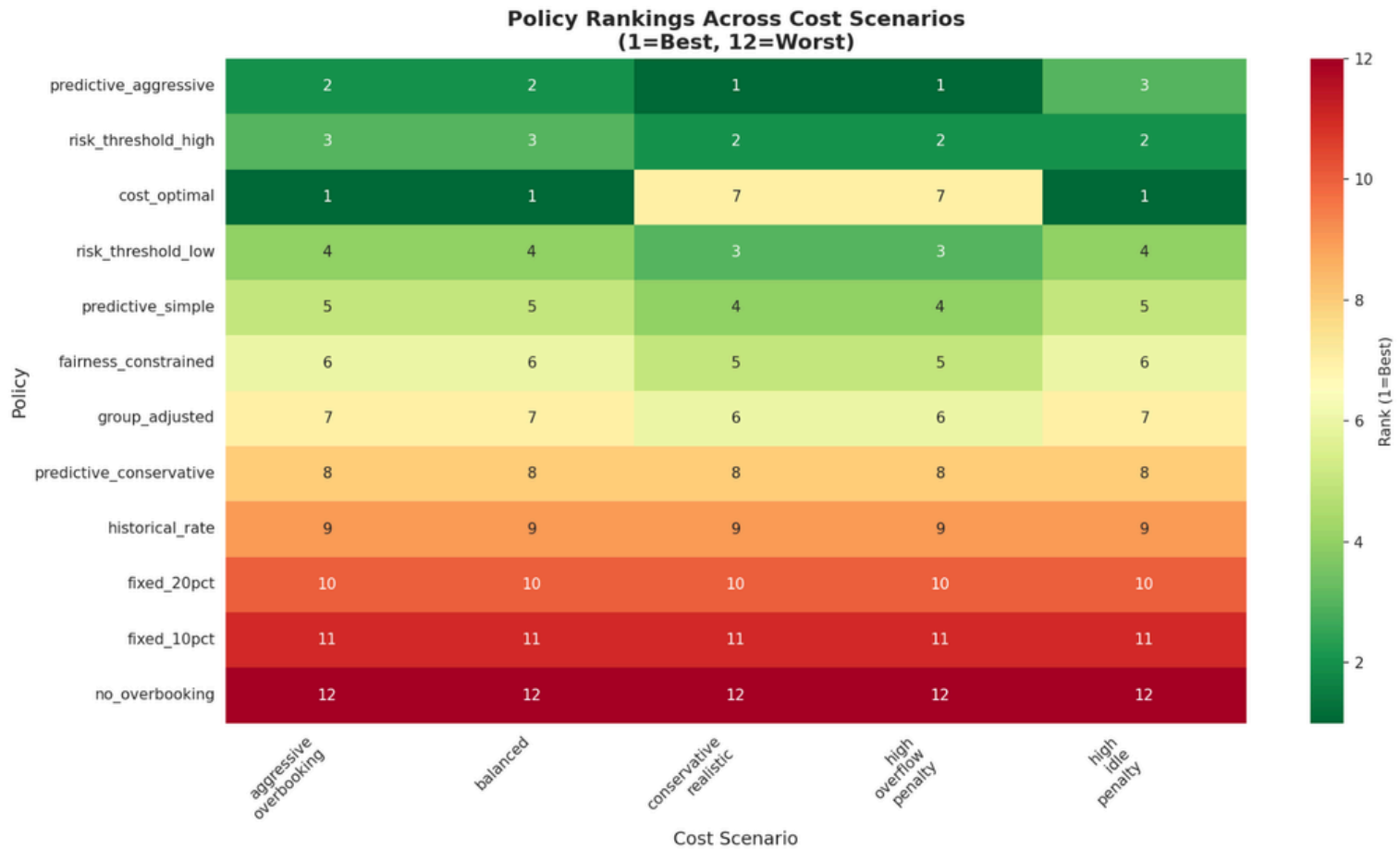
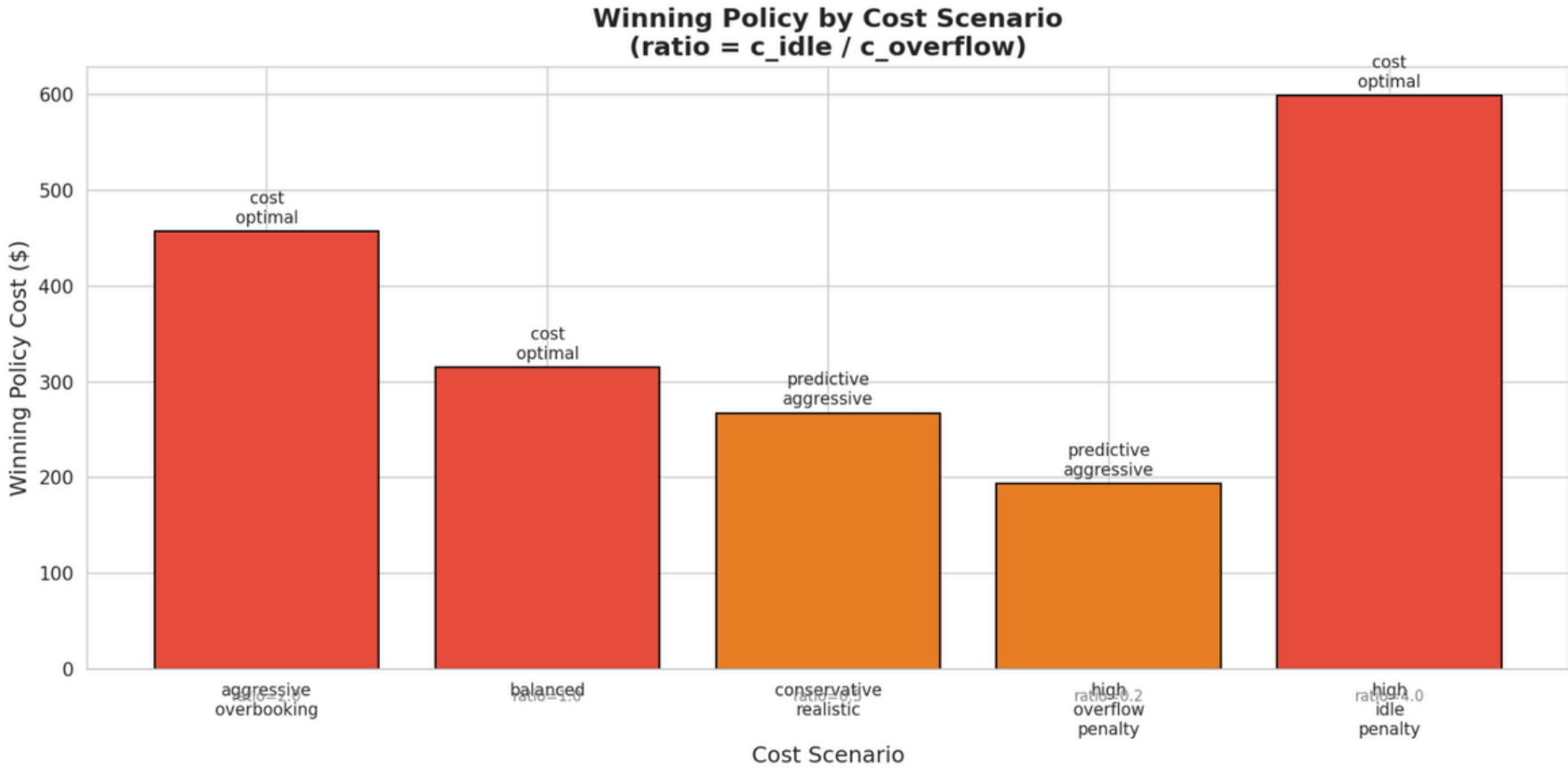
Primary Finding: ML-Based Overbooking Saves 64.5%

| Policy                | Mean Cost | Savings vs Baseline | Utilization |
|-----------------------|-----------|---------------------|-------------|
| cost_optimal          | \$457     | 64.50%              | 85.30%      |
| predictive_aggressive | \$500     | 61.20%              | 83.70%      |
| predictive_simple     | \$576     | 55.30%              | 81.00%      |
| no_overbooking        | \$1,289   | —                   | 57.00%      |

Critical Insight: Winner Depends on Cost Structure

| If clinic values...      | Cost Ratio                                | Winner                |
|--------------------------|-------------------------------------------|-----------------------|
| Revenue (idle expensive) | $c_{\text{idle}}/c_{\text{overflow}} > 1$ | cost_optimal          |
| Patient satisfaction     | $c_{\text{idle}}/c_{\text{overflow}} < 1$ | predictive_aggressive |

Implication: Clinics should assess their cost structure before choosing a policy - there's no universal "best" answer.



# Key Insights - Robustness & Fairness Insights

## Robustness Finding: cost\_optimal is Most Robust

| Policy            | Robustness Score | Cost Range      |
|-------------------|------------------|-----------------|
| cost_optimal      | \$847            | \$205 - \$1,369 |
| predictive_simple | \$940            | \$217 - \$1,461 |
| no_overbooking    | \$1,532          | \$644 - \$1,920 |

## Key Insight: Model Degradation Has Minimal Impact

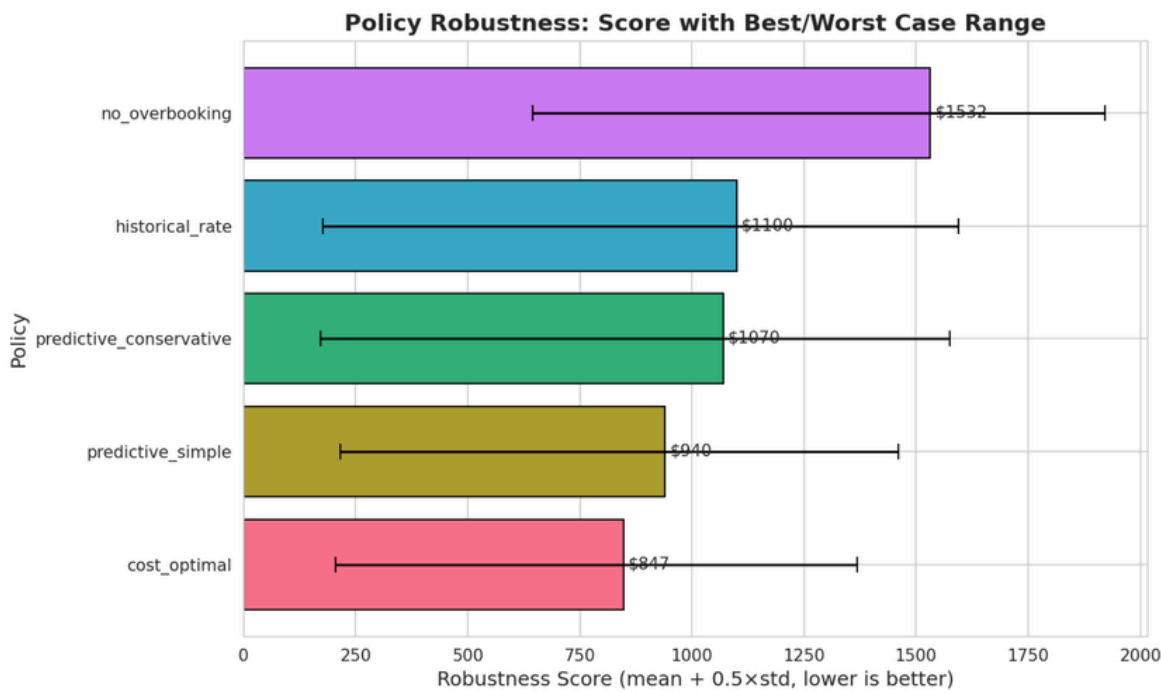
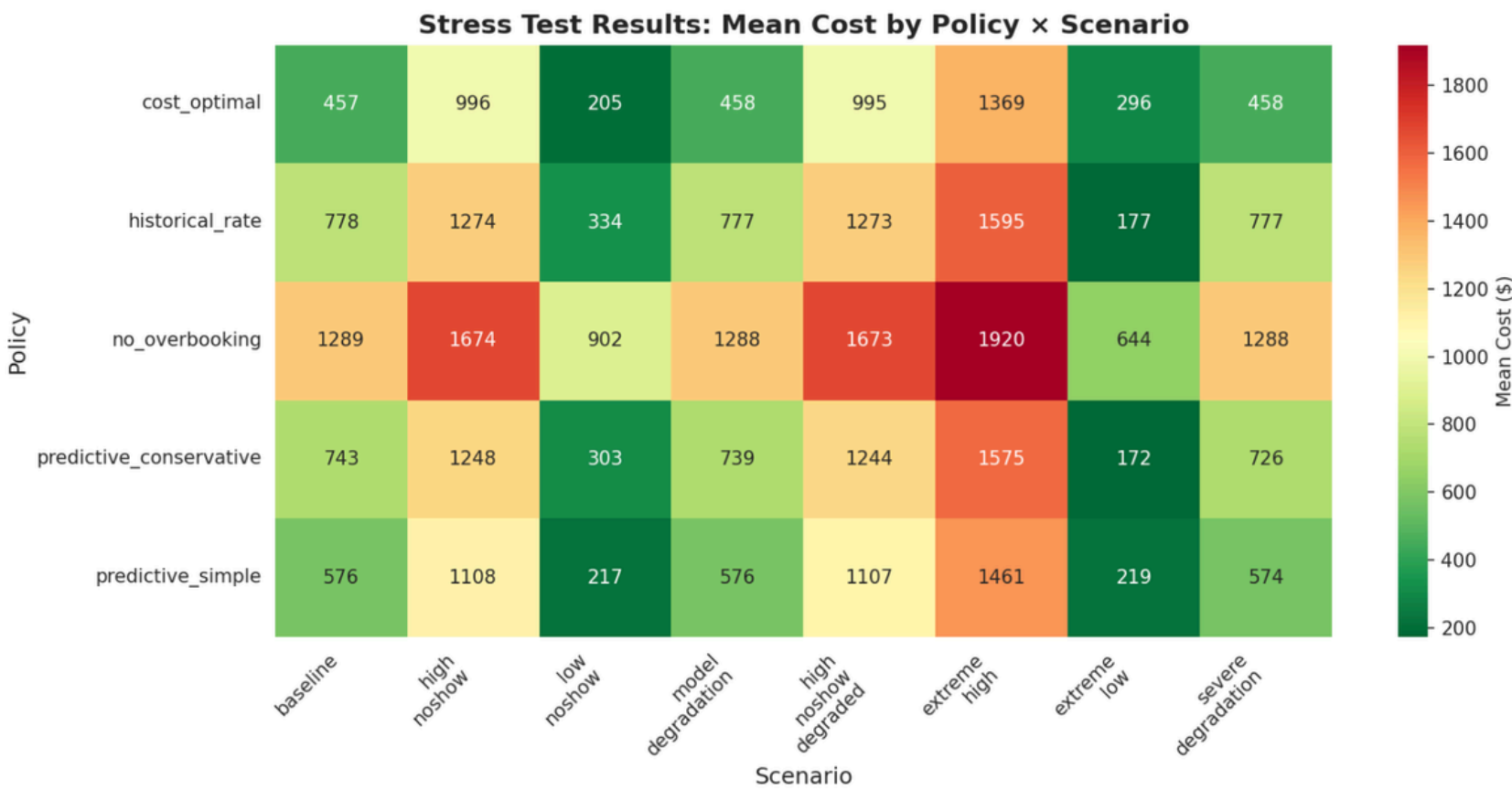
- 10% noise: <1% cost increase
- 20% noise: <1% cost increase
- Predictions are robust to drift!

## Surprising Fairness Result:

| Policy               | Combined Disparity | Cost           |
|----------------------|--------------------|----------------|
| predictive_simple    | 0.37% (fairest)    | \$576 (lowest) |
| fairness_constrained | 0.47%              | \$589          |
| group_adjusted       | 0.47%              | \$638          |

Key Takeaway: Simple ML-based policies can achieve BOTH lower cost AND better fairness than explicitly "fairness-constrained" approaches.

"Fairness-constrained" policy is actually LESS fair than predictive\_simple!  
→ No cost-fairness tradeoff needed





# Major Conclusions and Challenges

✓ **Approach Viability: CONFIRMED**

- 64.5% cost reduction achievable (\$832/slot, \$598K annually)
- Statistically significant ( $p < 0.0001$ )
- Robust across 8 stress scenarios

 **Unexpected Results:**

- fairness\_constrained is LESS fair than predictive\_simple
- Model noise has negligible impact on costs (<1%)
- predictive\_aggressive more robust than cost\_optimal across cost scenarios (avg rank 1.8 vs 3.4)

 **Difficulties Encountered:**

| Issue                    | Description                                       | Impact                                                                                                      |
|--------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Model Calibration Gap    | Predicted 42.9% vs actual 26.1% no-show rate      | Inflates k values; relative comparisons/policy rankings still valid since all policies are affected equally |
| Gradient Descent Failure | Stuck at k=0 due to non-convex region at low k    | Enumeration used as fallback                                                                                |
| Cost Function Mismatch   | Optimization vs Simulation use different formulas | Results internally consistent but not directly comparable                                                   |

# Suggestions for Future Work and Improvements

## Technical Improvements:

| Priority | Improvement                                                                 | Expected Impact                                 |
|----------|-----------------------------------------------------------------------------|-------------------------------------------------|
| HIGH     | <b>Fix model calibration</b> (Platt scaling / isotonic regression)          | Align predicted probabilities with actual rates |
| HIGH     | <b>Incorporate additional features</b> (weather, day-of-week, SMS response) | Improve prediction accuracy?                    |
| MEDIUM   | <b>Multi-objective optimization</b> (cost + fairness + robustness)          | Explicit Pareto-optimal solutions               |

-  **If unlimited resources:**
- Partner with healthcare system for full-scale implementation
  - Develop patient-facing app with appointment reminders
  - Measure actual ROI and patient satisfaction impact

## Implementation Path:

- Pilot Study: A/B test with real clinic to validate simulation
- Real-Time Deployment: API integration with scheduling system
- Dynamic Re-calibration: Monitor drift and retrain automatically

## Extended Analysis:

- Larger dataset (multiple clinics, longer time horizon)
- Patient-level outcome tracking (did overflow patients return?)
- Cost estimation refinement (survey actual clinic costs)

# References

1. Kheirkhah, P., et al. (2016). "Prevalence, predictors and economic consequences of no-shows." *BMC Health Services Research*, 16:13.
2. Dantas, L.F., et al. (2018). "No-shows in appointment scheduling – a systematic literature review." *Health Policy*, 122(4), 412-421.